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National Institute of Technology Goa

Programme Name: B.Tech.

Mid Semester Examinations, March 2026

Course Name: Computer Programming and Problem Solving

Date: 5/03/2026

Duration: 1 Hour 30 Minutes

Course Code: CS100

Time: 10:00 AM - 11:30 AM

Max. Marks: 50

ANSWER ALL QUESTIONS

Note: Please provide justifications if you believe any program contains error(s).

1. (a) What will be the output of the following programs?

```
i. #include <stdio.h>
int main()
{
    int i = 0, n = 0;
    while(i<=n)
    {
        if(i%2) i++; break;
        i++;
    }
    printf("%d", i);
    return 0;
}
```

[3]

```
ii. #include <stdio.h>
int main()
{
    int i = 5;
    do {
        printf("%d ", i);
        i--;
    } while(i >= 5);
    printf("%d", i);
    return 0;
}
```

[3]

(b) Write a C program to determine whether a triangle can be formed using three given angles. If the sum of the angles is not equal to 180° , it is not valid.

Example:

Input: 40 55 65

Output: Not valid

[4]

2. (a) What will be the output of the following programs?

```
i. #include <stdio.h>
int main()
{ int a, b;
  a = 1;
  b = 2;
  switch (a) {
    case 1:
      switch (b) {
        case 2:
          printf("Statement 1");
          break;
        default:
          printf("Statement 2");
      }
      break;
    default:
      printf("Statement 3");
  }
  return 0;
}
```

[3]

```
ii. #include <stdio.h>
int main()
{
  int a = 4, n = 2, i, result = 0;
  for (i = 0; i < n; i++)
    result += a;
  printf("%d", result);
  return 0;
}
```

[3]

(b) Write a C program to display the characters from A to Z using for loop.

[4]

3. (a) What will be the output of the following programs?

```
i. #include <stdio.h>
int main() {
  char str[] = "NIT GOA";
  printf("%c", str[3]);
  return 0;
}
```

[2]

```

ii. #include <stdio.h>
#include <string.h>
int main() {
    char str[20] = "PROGRAMMING";
    printf("%c", str[0]);
    str[1] = 'r';
    printf("%c", str);
    return 0; }

```

[3]

- (b) Write a C program to compute the sum of the diagonals of a square matrix. The program should calculate: (i) the sum of the primary diagonal (top-left to bottom-right), and (ii) the sum of the secondary diagonal (top-right to bottom-left).

Example

Input:

```

1 2 3
4 5 6
7 8 9

```

Output:

```

Sum of primary diagonal elements = 15
Sum of secondary diagonal elements = 15

```

[5]

4. (a) What will be the output of the following programs?

```

i. #include <stdio.h>
int main()
{
    int arr[] = {1, 2, 3, 4, 5};
    int i = 0;
    printf("%d\n", arr[i++]);
    printf("%d\n", arr[++i]);
    printf("%d\n", arr[i]);
    return 0; }

```

[3]

```

ii. #include <stdio.h>
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    int i;
    for(i = 0; i < 5; i++) {
        arr[i] = arr[i] + i;
    }
    for(i = 0; i < 3; i++) {
        printf("%d ", 2*arr[i]);
    }
    return 0; }

```

[3]

(b) Write a C program that finds the smallest element in an array and prints all the indices where it occurs.

[4]

5. (a) i. In C programming, what is the order of evaluation for the following expression?

$a = w \% x / y * z;$

[2]

ii. What will be the output of the following program?

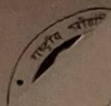
```
#include <stdio.h>
int main()
{
    int a = 4, b = 3, c = 2;
    int result;
    result = a - b + c / 3;
    printf("%d", result);
    return 0;
}
```

[3]

(b) Write a C program to multiply two floating-point numbers using a function that takes arguments and returns a value.

[5]

ALL THE BEST



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NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech

End Semester Examinations, December-2025

Course Name: Computer Programming and Problem Solving

Date: 13/12/2025

Duration: 3 Hours

Course Code: CS100

Time: 10:00 AM-01:00 PM

Max. Marks: 100

ANSWER ALL QUESTIONS

NOTE: There are 8 main questions and 4 pages in this question paper.

1.

- a. Determine which of the following are valid variable names (identifiers)? If invalid, explain why
- i. 1record
 - ii. file_3
 - iii. return
 - iv. name and address
 - v. 123_3_10
 - vi. first.file
- b. Determine which among the following are valid string constants. In case of invalid, give reason for the same.
- i. "Red, White and Blue"
 - ii. "Name :
 - iii. 'Chap. 3 (Const\ 'd) "
 - iv. "The Professor said, "Please don't sleep in class"
- c. A program written in C contains the following declarations
- ```
int i, j;
float x;
char c;
```
- Determine the data type of the following expressions
- i. i+c
  - ii. x+c
  - iii. i+x
  - iv. j+x
- d. What is the value resulting in the following expressions showing each step of evaluating the expression
- i.  $a = 2 * 3 / 4 + 4 / 4 + 8 - 2 + 5 / 8$ ;
  - ii.  $a = b = c = 3 + 4$ ;
  - iii.  $x = 2 - 3 + 5 * 2 / 8 \% 3$
  - iv.  $y = z = -3 \% -8 / 2 + 7$ ;

[3+2+2+8=15 marks]

2.

- a. A program written in C contains the following declarations and initial assignments
- ```
int i = 8, j = 5, k;  
float x = 0.005, y = -0.01 ;  
char c = 'c', d = 'd';
```
- Determine the value of each of the following expressions that is assigned to k. Use the value initially assigned to the variables for each expression
- i. $k = 2 * ((i / 5) + (4 * (j - 3))) \% (i + j - 2)$;
 - ii. $k = (x > y) \&\& (i > 0) \&\& (j < 5)$;
 - iii. $k = (j == 5) ? i : j$;

b. Write a switch-case statement that will examine the value of an integer variable flag and print out the following messages depending on the value assigned to flag.

- "HOT" if flag has value 1
- "LUKE WARM" if flag has value 2
- "COLD" if flag has value 3
- "OUT OF RANGE" if flag has a value other than above

[6+4=10 marks]

3. A C program contains the following statements

```
float a=0.001, b=0.003;  
float c, *pa, *pb, *pc;
```

```
pa=&a;  
*pa=2*a;  
pb=&b;  
c=3*(*pb-*pa);  
pc=&c;  
pc--;
```

Suppose each floating-point number occupies 4 bytes of memory. If the value assigned to **a** begins at (hexadecimal) address 1130, the value assigned to **b** begins at (hexadecimal) address 1134, and the value assigned to **c** begins at (hexadecimal) address 1136, then

- What value corresponds to &a?
- What value corresponds to &b?
- What value corresponds to &c?
- What value is assigned to pa?
- What value is represented by *pa?
- What value is represented by &>(*pa)?
- What value is assigned to pb?
- What value is represented by *pb?
- What value is assigned to c?
- What value is assigned to pc?

[10 marks]

4. Write down the output of each of the following programs. If there is any error, clearly explain what is the error.

```
a. #include <stdio.h>  
int main()  
{  
    int i=0, x=1;  
    for (i=1; i<10; ++i)  
    {  
        if ( i % 2 == 1)  
            x+=i;  
        else  
            x--;  
        printf("%d ", x);  
    }  
    printf("\n%d ", x);  
    return 0;  
}
```

```

b. #include <stdio.h>
int main()
{
    int a, b=0;
    int c[10]={2,1,4,3,5,6,7,9, 10,0};

    for (a=0; a<10;a++)
    {
        if(a%2==0)
            b+=c[a];
    }
    printf("%d\n", b);
    return 0;
}

```

```

c. #include <stdio.h>
void sub1(int x[][4], int, int);
int main()
{
    int z[3][4]={1,2,3,4,5,6,7,8,9,10,11,12};
    sub1(z, 3,4);
    return 0;
}
void sub1(int x[][4], int rows, int cols)
{
    int a, b, c;
    for (b=0;b<cols; b++)
    {
        c=0;
        for (a=0;a<rows;a++)
        {
            if(x[a][b]>c)
                c=x[a][b];
        }
    }
    printf("%d ", c);
    return 0;
}

```

```

d. #include <stdio.h>
int main(){
    FILE *fp;
    char ch;
    int count1=0, count2=0;
    fp=fopen("data.txt ", "r ");
    while(1)
    {
        ch=fgetc(fp);
        if(ch==EOF)
            break;
        count1++;
        if(ch== ' ')
            count2++;
    }
    fclose(fp);
    printf("Count1=%d\n", count1);
    printf("Count2=%d\n", count2);
    return 0;
}

```

Content of data.txt file is given in the following box:

The National Institute of Technology Goa (NIT Goa) is a premier national-level technical institute in India established in 2010 by an act of parliament (NIT Act, 2007 and NIT (Amendment) Act, 2012). NIT Goa is an autonomous institute functioning under the aegis of Ministry of Education (MoE), Government of India, and has been declared an "Institute of National Importance".

[2.5x4=10 marks]

5. Explain the bit operations carried out using ~, >>, <<, & and | giving suitable examples [10 marks]
6. Write a C program to generate the following pyramid of numbers. The number of rows must be taken from the user.

```

      1
     2 2
    3 3 3
   4 4 4 4
  5 5 5 5 5

```

[15 marks]

7. Write a C program that sorts a given array of integers in ascending order. The elements of the array must be taken from the user. [Ex: If the array is: 10, 20, 1, 4, 30, 6, 7, the sorted array will be: 1, 4, 6, 7, 10, 20, 30].

[15 marks]

8. Write a C Program that reads in a string from user. The string may be a multi word string. Upon reading the string, the program gives the following options to the user.
 - a. Count the number of words in the string
 - b. Check whether the string is a palindrome. A palindrome is a string that reads the same from front as well as back. (Ex: MALAYALAM)
 - c. Reverse the string

For each of the tasks in a, b, and c write a function which will be called based on user choice. NOTE: You are not supposed to use any built in string functions for the implementation for options a to c.

[15 marks]

*****ALL The Best*****

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National Institute of Technology Goa

Programme Name: B.Tech.

End Semester Examinations, May 2026

Course Name: Computer Programming and Problem Solving

Date: 04/05/2026

Duration: 3 Hours

Course Code: CS100

Time: 10:00 AM - 1:00 PM

Max. Marks: 100

ANSWER ALL QUESTIONS

Note: Please provide justifications if you believe any program contains error(s).

1. (a) What will be the output of the following programs?

```
i. #include <stdio.h>
int main() {
    int a = 6;
    printf("%d %d %d", a, a++, ++a);
    return 0; }
```

[3]

```
ii. #include <stdio.h>
int main() {
    int i = 0;
    for (; i < 3; printf("%d ", i))
        i++;
    return 0; }
```

[3]

(b) Write a C program to swap two numbers without using a third variable.

[4]

2. (a) What will be the output of the following programs?

```
i. #include <stdio.h>
int main()
{ int i = 0;
  do {
    if (i == 4)
      i++;
  } while (i < 10)
  printf("%d\n", i);
  i++;
  return 0; }
```

[2]

```

ii. #include<stdio.h>
int print(int n) {
    if (n == 0)
        return 1;
    printf("%d", n);
    print(n-1);
}
int main()
{
    printf("%d", print(2));
    return 0;
}

```

[3]

(b) Write a C program to find the factorial of a number using (a) normal loop and (b) recursion. [5]

3. (a) What will be the output of the following programs?

```

i. #include<stdio.h>
int main()
{
    int i;
    for (i = 0; i < 6; i++) {
        if (i == 2) {
            continue;
        }
        if (i == 4) {
            break;
        }
        printf("%d\n", i);
    }
    return 0;
}

```

[2]

```

ii. #include<stdio.h>
void addOne(int* ptr) {
    (*ptr)++;
}
int main()
{
    int* p, i = 10;
    p = &i;
    addOne(p);
    printf("%d%d", *p+1, (*p)++);
    return 0;
}

```

[3]

(b) Write a C program to compute the following:

$$1 - 2 + 3 - 4 + 5 - 6 + \dots \pm n$$

Example:

Input: $n = 5$

Calculation: $1 - 2 + 3 - 4 + 5 = 3$

Output: 3

[5]

4. (a) What will be the output of the following programs?

```
i. #include <stdio.h>
int main() {
    int a = 10;
    int *p = &a;
    int **q = &p;
    printf("%d", **q);
    return 0; }
```

[2]

```
ii. #include <stdio.h>
int main() {
    int arr[] = {1, 2, 3};
    printf("%d", *(arr + *(arr + 1)));
    return 0; }
```

[3]

[5]

(b) Write a C program to separate even and odd numbers into two arrays.

5. (a) What will be the output of the following programs?

```
i. #include <stdio.h>
int main() {
    int arr[] = {1, 2, 3, 4};
    int *p = arr + 1;
    printf("%d", *(p++));
    printf("%d", *p);
    return 0; }
```

[2]

```
ii. #include <stdio.h>
int main() {
    int x = 10;
    printf("%d", x > 10 ? 100 : 200);
    return 0;
}
```

[3]

- (b) Write a C program to find the missing number in an array of size $n - 1$, where the array contains distinct integers (unsorted) ranging from 1 to n . Do not use any mathematical formula (such as $n(n + 1)/2$). [5]

6. (a) What will be the output of the following programs?

```
i. #include <stdio.h>
void fun(int a) {
    a = a + 10;
}
int main() {
    int a = 5;
    fun(a);
    printf("%d", a);
    return 0; }
```

[2]

```
ii. #include <stdio.h>
int sample() {
    return 10;
}
int main() {
    printf("%d", sample() + sample());
    return 0; }
```

[3]

- (b) Write a C program to find the largest element in an array using functions. The program should accept n elements from the user and store them in an array. Create a user-defined function that takes the array and its size as arguments and returns a pointer to the largest element present in the array. In the main() function, use this returned pointer to display the largest element.

Example:

Enter number of elements: 5

Enter elements: 10 25 3 30 16

Largest element is 30

[5]

7. (a) What will be the output of the following programs?

```
i. #include<stdio.h>
#include<string.h>
int main()
{
    char a[] = "hello";
    char b[] = "hello";
    if (a == b) {
        printf("Equal\n");
    }
    printf("End");
    return 0; }
```

[2]

```

ii. #include<stdio.h>
    #include<string.h>
    int main()
    {   char p[20];   char *s = "string";
        int length = strlen(s);
        int i;
        for (i = 0; i < length; i++)
            p[i] = s[length - i];
        printf("%s", p);
        return 0;   }

```

[3]

- (b) Write a C program to count the number of digits in a given integer using a user-defined function. The program should include a main() function where the user inputs an integer value. A separate function should be created that takes this integer as an argument and returns the number of digits present in it. The result should be displayed in the main() function after calling the user-defined function.

Example:

Enter a number: 435620

Number of digits: 6

[5]

8. (a) What will be the output of the following programs?

```

i. #include <stdio.h>
    union Job1
    {   char name[20];   float salary;   int workerNo;
    }uJob;
    struct Job2
    {   char name[30];   int workerNo;
    }sJob;
    int main()
    {   printf("%d bytes", sizeof(uJob));
        printf("\n%d bytes", sizeof(sJob));
        return 0;   }

```

[2]

```

ii. #include <stdio.h>
    union Job{
        float salary;   int workerNo;
    };
    int main(){
        j.salary = 5.3;   j.workerNo = 20;
        printf("Salary = %.1f\n", j.salary);
        printf("WorkerNo = %d", j.workerNo);
        return 0;   }

```

[3]

- (b) Write a C program to store the details of n employees using an array of structures, where each employee has an ID, name, and salary. The program should display the details of all employees, find and display the employee with the highest salary, should calculate the average salary of all employees and count how many employees have a salary above the average. [5]

9. (a) What will be the output of the following programs?

```
i. #include <stdio.h>
#include <stdlib.h>
int main() {
    int *ptr = (int*) malloc(5 * sizeof(int));
    int *p = ptr;
    for(int i = 0; i < 5; i++) {
        *p = i + 10;
        p++;
    }
    p = ptr;
    for(int i = 0; i < 5; i++) {
        printf("%d ", *p);
        p++;
    }
    free(ptr);
    return 0; }
```

[2]

```
ii. #include <stdio.h>
#define LOOP(n) \
    for(int i = 0; i < n; i++) \
        printf("%d ", i);
int main() {
    LOOP(3 + 2); return 0; }
```

[3]

- (b) Write a C program using macros to implement a simple calculator that performs addition, subtraction, multiplication, and division of two numbers. Define separate macros for each operation. The program should take two integer inputs from the user and ask the user to choose the operation using a menu. Based on the user's choice, the corresponding macro should be used to display the result. [5]

10. (a) Perform the following conversions between number systems. Show all the steps involved in the conversion process.

i. Convert the $(2C4)_{16}$ to decimal number system. [2]

ii. Convert $(111010011)_2$ to hexadecimal number system. [3]

- (b) Write a C program to create a file and store details of n students (ID, name, and marks) using structures. The program should write the student data into a file and then read the contents of the file to display all the stored records. [5]